

Stochastic Option Pricing and Model Risk Analysis

Black–Scholes Model, Brownian Motion, Monte Carlo Simulation, and Greeks on SPY Options

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Abstract

In this project, I build an end-to-end stochastic option pricing framework for European options written on the SPDR S&P 500 ETF Trust, commonly known as SPY. I begin by defining the payoff structure of a European call option and then model the underlying asset price using Brownian Motion and Geometric Brownian Motion. I then implement the Black–Scholes analytical pricing formula, validate the result through Monte Carlo simulation, compute Greeks to study risk sensitivities, and extend the analysis toward historical volatility calibration, implied volatility, and model risk diagnostics. The goal of this project is to connect stochastic calculus, derivative pricing, numerical simulation, and model validation in a structured quantitative finance workflow.

The full Python implementation, generated figures, and project files are available in the GitHub repository: [stochastic-option-pricing-spy](https://github.com/sounakdas/stochastic-option-pricing-spy).

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1 Project Objective

The objective of this project is to build a stochastic option pricing and model validation framework for European options written on SPY. The project is designed to demonstrate the complete workflow of a derivatives pricing model:

- define the option pricing problem,
- model the underlying asset using stochastic processes,
- implement Black–Scholes analytical pricing,
- validate the analytical price using Monte Carlo simulation,
- compute Greeks for risk sensitivity,
- estimate volatility from historical data,
- examine implied volatility and model risk.

The underlying asset is SPY, an exchange-traded fund tracking the S&P 500 Index. SPY is chosen because it is highly liquid and has actively traded options.

2 European Call Option Payoff

A European call option gives the holder the right, but not the obligation, to buy the underlying asset at a fixed strike price K at maturity T .

The payoff at maturity is:

$$\max(S_T - K, 0),$$

where S_T is the price of SPY at maturity.

If $S_T > K$, the option finishes in the money and has positive payoff. If $S_T \leq K$, the option expires worthless.

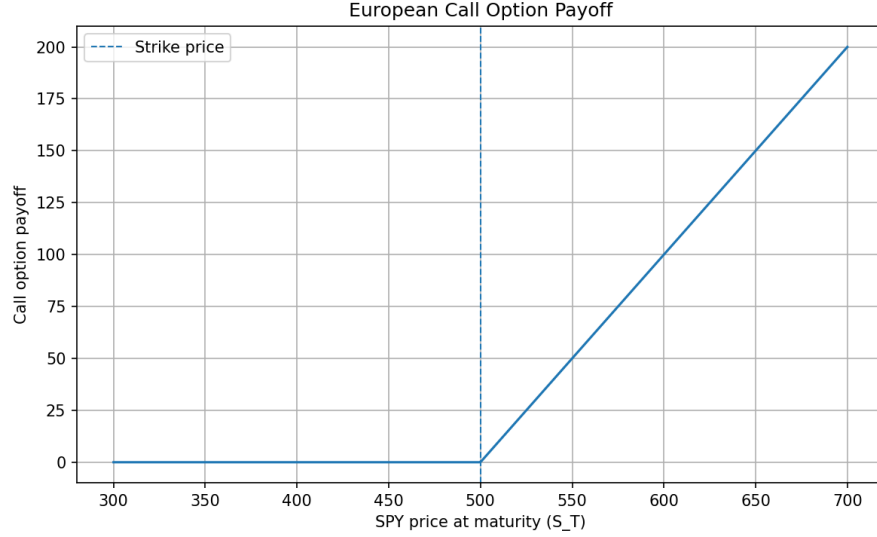


Figure 1: European call option payoff. The payoff is zero below the strike and increases linearly once the terminal SPY price exceeds the strike.

Figure 1 shows the nonlinear nature of option payoffs. This nonlinearity is the reason volatility becomes so important in option pricing.

3 Stochastic Modeling of SPY Price

The future price S_T is unknown today. Therefore, the underlying asset price must be modeled as a stochastic process.

The Black–Scholes framework assumes that the asset follows Geometric Brownian Motion:

$$dS_t = \mu S_t dt + \sigma S_t dW_t.$$

Here:

- μ is the expected return,
- σ is volatility,
- W_t is Brownian Motion,
- dW_t represents a random market shock.

Brownian Motion is the source of randomness in the model. It has continuous paths and normally distributed independent increments.

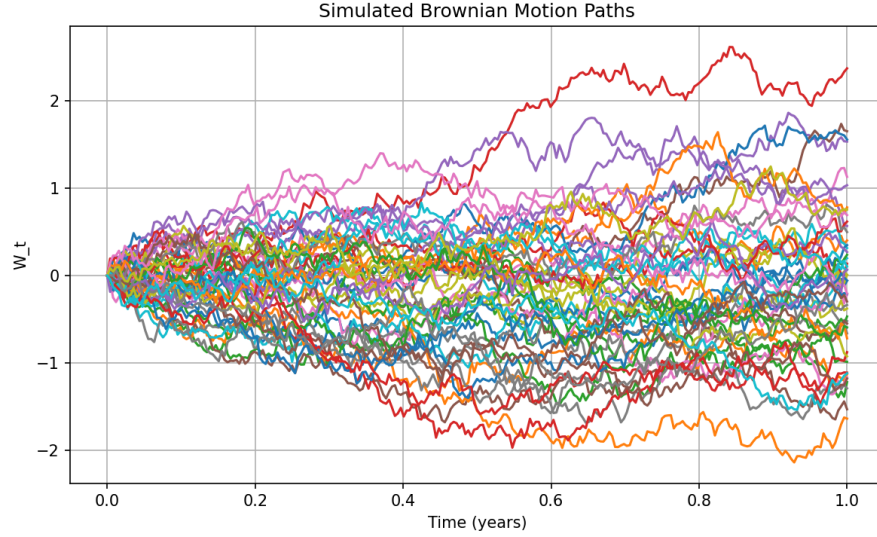


Figure 2: Simulated Brownian Motion paths. These paths represent the continuous random shocks driving the asset price model.

The Geometric Brownian Motion solution is:

$$S_t = S_0 \exp \left[\left(\mu - \frac{1}{2} \sigma^2 \right) t + \sigma W_t \right].$$

This exponential form ensures that $S_t > 0$, which is desirable for equity and ETF prices.

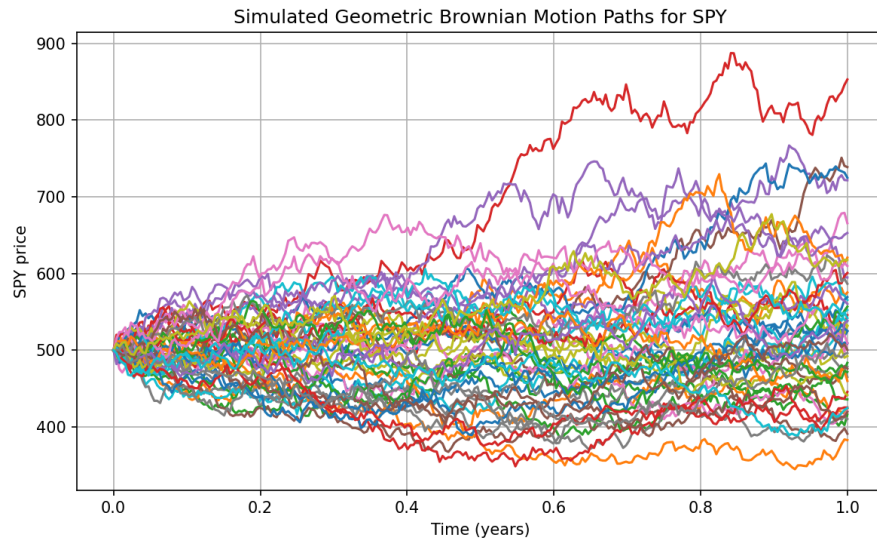


Figure 3: Simulated Geometric Brownian Motion paths for SPY. The paths remain positive and show different possible future price scenarios.

4 Black–Scholes Analytical Pricing

The Black–Scholes formula for a European call option is:

$$C_0 = S_0 N(d_1) - K e^{-rT} N(d_2),$$

where:

$$d_1 = \frac{\ln(S_0/K) + (r + \frac{1}{2}\sigma^2) T}{\sigma\sqrt{T}},$$

and:

$$d_2 = d_1 - \sigma\sqrt{T}.$$

The European put price is:

$$P_0 = K e^{-rT} N(-d_2) - S_0 N(-d_1).$$

The project used the following base parameters:

Parameter	Value
Current SPY price S_0	500
Strike price K	500
Time to maturity T	1 year
Risk-free rate r	5%
Volatility σ	20%

Using these parameters, the implemented Black–Scholes model produced:

Quantity	Value
Call price	52.252918
Put price	27.867630
d_1	0.350000
d_2	0.150000

The put-call parity condition was also verified:

$$C_0 - P_0 = S_0 - K e^{-rT}.$$

This confirms the internal consistency of the implementation.

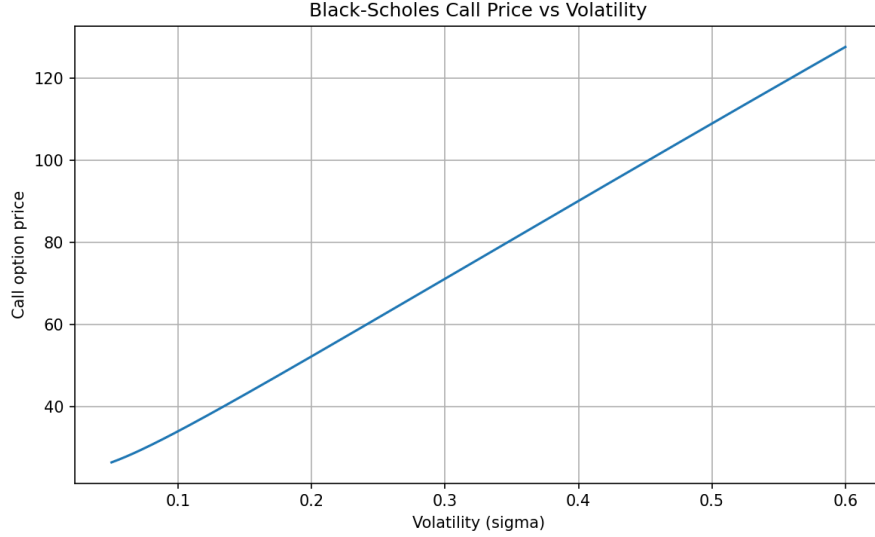


Figure 4: Black–Scholes call price as a function of volatility. The call price increases with volatility because the upside is amplified while the downside payoff is bounded below by zero.

Figure 4 highlights one of the most important observations of the project: volatility is a dominant input in option pricing.

5 Monte Carlo Validation

Under the risk-neutral measure, the terminal price is simulated as:

$$S_T = S_0 \exp \left[\left(r - \frac{1}{2} \sigma^2 \right) T + \sigma \sqrt{T} Z \right],$$

where:

$$Z \sim N(0, 1).$$

The Monte Carlo estimator for a European call option is:

$$\hat{C}_0 = e^{-rT} \frac{1}{M} \sum_{i=1}^M \max(S_T^{(i)} - K, 0).$$

Using 100,000 simulations, the project obtained:

Quantity	Value
Black–Scholes price	52.252918
Monte Carlo price	52.369460
Standard error	0.232947
95% CI lower bound	51.912885
95% CI upper bound	52.826035
Absolute difference	0.116542

The Black–Scholes analytical price lies inside the Monte Carlo 95% confidence interval. This validates the Monte Carlo implementation.

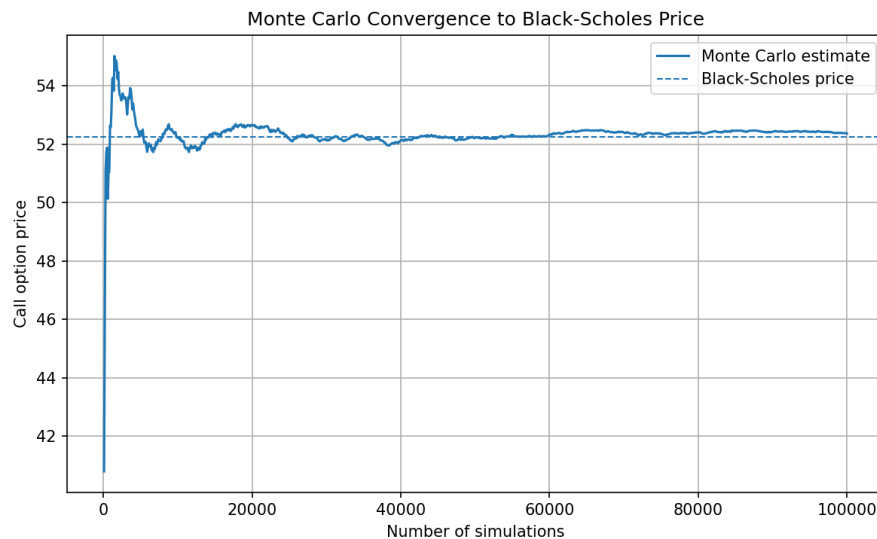


Figure 5: Monte Carlo convergence to the Black–Scholes analytical price. As the number of simulations increases, the Monte Carlo estimator stabilizes around the closed-form price.

Figure 5 demonstrates that simulation error decreases as the number of simulations increases. This is consistent with the law of large numbers.

6 Greeks and Risk Sensitivities

Greeks measure how sensitive the option price is to different inputs. They are important for hedging, risk management, and model validation.

The project computed the following Greeks:

Greek	Value
Call Delta	0.636831
Put Delta	-0.363169
Gamma	0.00375240
Vega	187.620173
Call Theta per year	-32.070138
Put Theta per year	-8.289402
Call Rho	266.162408
Put Rho	-209.452305

In market convention:

- Vega per 1 volatility point is approximately 1.8762.
- Call theta per day is approximately -0.1273 .
- Call rho per 1 rate point is approximately 2.6616.

6.1 Delta

Delta measures directional exposure to the underlying asset:

$$\Delta = \frac{\partial C}{\partial S}.$$

For a call option:

$$\Delta_{\text{call}} = N(d_1).$$



Figure 6: Delta versus underlying SPY price. Call delta increases as the option moves deeper into the money.

6.2 Gamma

Gamma measures the curvature of the option price:

$$\Gamma = \frac{\partial^2 C}{\partial S^2}.$$

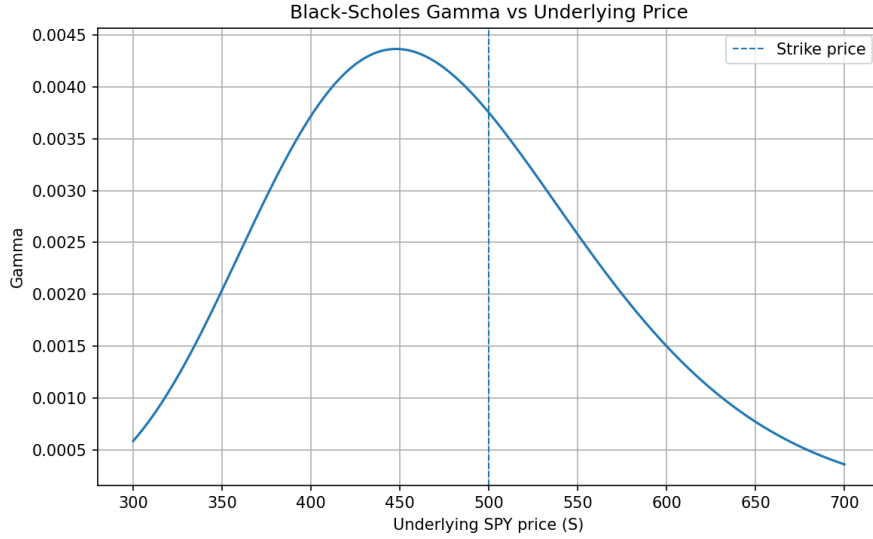


Figure 7: Gamma versus underlying SPY price. Gamma is highest near the strike, where delta changes most rapidly.

6.3 Vega

Vega measures sensitivity to volatility:

$$\text{Vega} = \frac{\partial C}{\partial \sigma}.$$

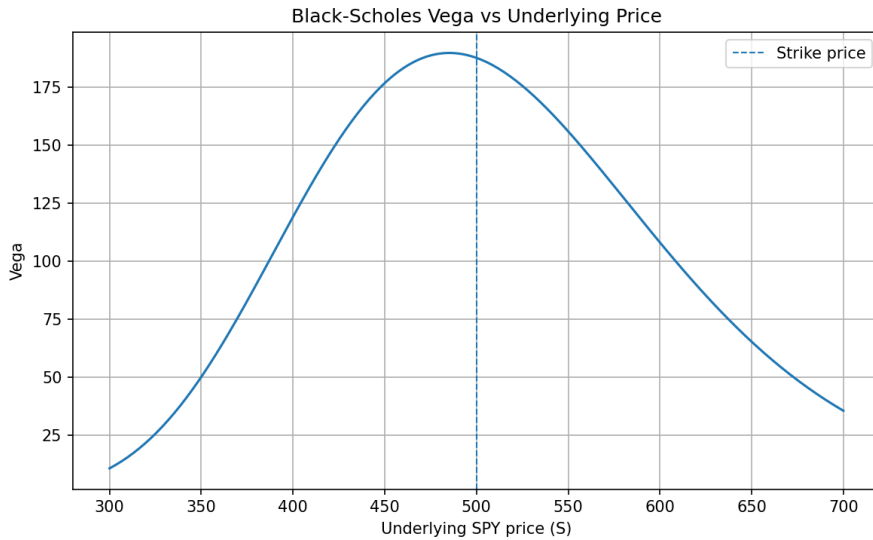


Figure 8: Vega versus underlying SPY price. Vega is highest around the at-the-money region.

These Greeks show that option risk is not only directional. It also depends heavily on curvature, volatility, time decay, and interest rate sensitivity.

7 Historical Volatility and Model Risk Diagnostics

The Black–Scholes model assumes constant volatility. To examine this assumption empirically, historical SPY log returns were computed as:

$$R_t = \log \left(\frac{S_t}{S_{t-1}} \right).$$

Daily volatility is estimated as:

$$\hat{\sigma}_{\text{daily}} = \text{Std}(R_t).$$

Annualized volatility is computed as:

$$\hat{\sigma}_{\text{annual}} = \hat{\sigma}_{\text{daily}} \sqrt{252}.$$

The rolling volatility plot is particularly important because it tests whether volatility is stable through time.

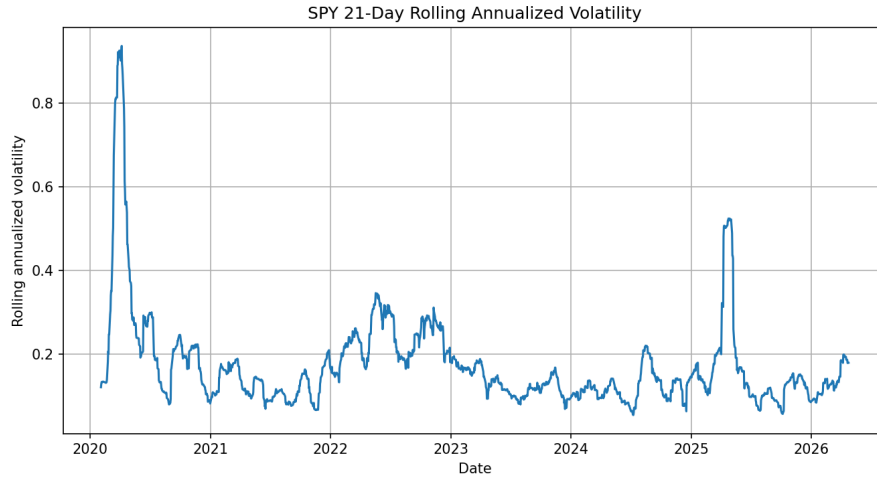


Figure 9: SPY 21-day rolling annualized volatility. The changing level of volatility shows that the constant volatility assumption is not fully realistic.

Figure 9 provides direct model risk evidence. If volatility changes through time, a single constant σ may not represent market conditions accurately.

8 Implied Volatility and Volatility Smile

Historical volatility is estimated from past returns. Implied volatility is backed out from market option prices.

Given a market call price C_{market} , implied volatility σ_{imp} solves:

$$C_{\text{BS}}(S_0, K, T, r, \sigma_{\text{imp}}) = C_{\text{market}}.$$

This project implemented implied volatility extraction using numerical root-finding.

If the Black–Scholes assumptions were perfectly correct, implied volatility would be approximately constant across strikes. In practice, implied volatility often varies by strike, producing a volatility smile or skew.

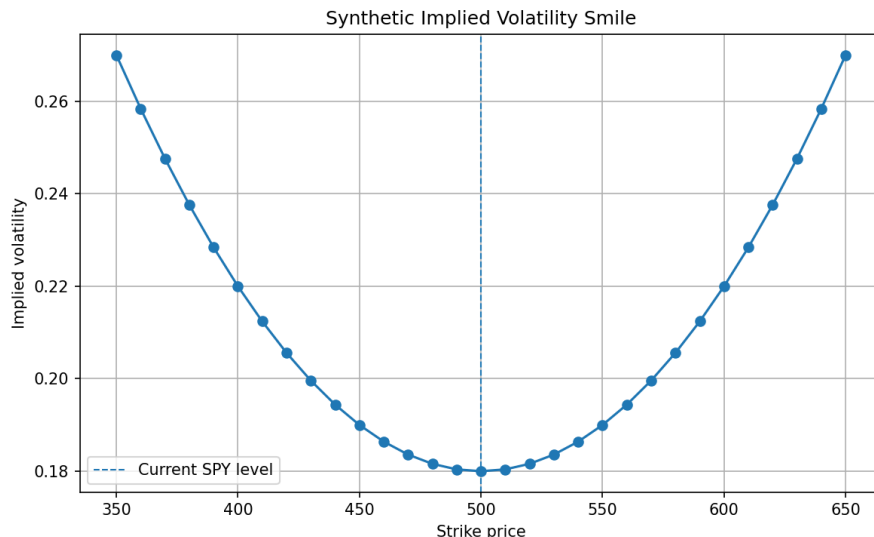


Figure 10: Synthetic implied volatility smile. The variation of implied volatility across strikes illustrates why the constant volatility assumption can be problematic.

The volatility smile is a key model risk diagnostic. It suggests that the market prices downside and upside risks differently from the assumptions of the basic Black–Scholes model.

9 Model Risk Discussion

The Black–Scholes model is elegant and analytically tractable, but it relies on strong assumptions.

Assumption	Real-World Issue	Model Risk
Constant volatility	Volatility changes through time	Mispricing risk
Lognormal prices	Returns may have fat tails	Tail risk understatement
Continuous trading	Markets close and trading is discrete	Hedging error
No transaction costs	Bid-ask spreads and costs exist	Hedge cost ignored
No jumps	Prices can gap after news	Jump risk ignored
Single volatility	Implied volatility varies by strike	Smile/skew not captured

From a model validation perspective, the important questions are:

- Is the GBM assumption reasonable for SPY?
- Is one constant volatility appropriate?
- How sensitive is the option price to volatility?
- Does Monte Carlo pricing validate the analytical price?

- Do Greeks behave as expected?
- What limitations should be disclosed before using the model?

10 Conclusion

This project built an end-to-end stochastic option pricing framework for SPY European options. The workflow started from the option payoff, moved to stochastic asset price modeling using Brownian Motion and Geometric Brownian Motion, implemented Black–Scholes analytical pricing, validated the result using Monte Carlo simulation, calculated Greeks, and finally examined volatility-related model risk.

The first major observation is that the European call option payoff is nonlinear and asymmetric. The payoff remains zero when the terminal SPY price is below the strike and increases linearly when the terminal price exceeds the strike. This justifies why volatility is valuable for option holders: higher volatility increases the probability of large upside outcomes, while the downside payoff of a long call is floored at zero.

The second observation is that Geometric Brownian Motion is a useful modeling choice because it keeps simulated asset prices positive. Brownian Motion itself can move above or below zero, but the exponential solution of GBM ensures positive prices. This is consistent with equity price modeling, although it is still an idealized assumption.

The third observation is that the Black–Scholes analytical price and the Monte Carlo price are very close. With the chosen parameters, the Black–Scholes call price was approximately 52.25, while the Monte Carlo price using 100,000 simulations was approximately 52.37. The analytical price also lay inside the Monte Carlo 95% confidence interval. This confirms that the simulation engine is correctly implementing the same risk-neutral pricing logic as the Black–Scholes formula.

A fourth observation is that the call price increases with volatility. This is theoretically justified because the call payoff is convex in the underlying price. More volatility increases the value of convex payoffs. Although it may initially appear counterintuitive that greater uncertainty increases option value, this happens because the long call benefits from upside movements while losses are limited to the premium paid.

The Greeks produced further important insights. Delta increased as the option moved deeper into the money, which is expected because the option starts behaving more like the underlying asset. Gamma was highest near the strike, showing that directional exposure changes most rapidly when the option is at-the-money. Vega was also highest near the at-the-money region, confirming that volatility matters most when there is significant uncertainty about whether the option will finish in the money. Theta was negative for the long call, reflecting time decay.

The main counterintuitive result is that the real-world drift μ does not appear in the Black–Scholes pricing formula. At first, this may seem surprising because one might expect the expected return of SPY to affect the option price. However, the Black–Scholes derivation constructs a locally risk-free hedged portfolio. Once the random component is hedged away, no-arbitrage requires the portfolio to earn the risk-free rate. Therefore, pricing is done under the risk-neutral measure, where the drift becomes r , not the real-world expected return μ .

The model risk diagnostics show that Black–Scholes is powerful but not fully realistic. The rolling volatility plot indicates that SPY volatility changes over time, which contradicts the constant volatility assumption. The implied volatility smile further suggests that the market does not price all strikes using one single volatility. These observations justify why Black–Scholes

is often used as a benchmark rather than a complete description of market behavior.

Overall, the project demonstrates both the strength and limitation of the Black–Scholes framework. It is mathematically elegant, internally consistent, and useful for understanding option pricing mechanics. However, practical use requires calibration, sensitivity analysis, and model risk awareness. This makes the project relevant not only for option pricing but also for model validation, market risk, and quantitative finance roles.

11 Implementation Details

The project was implemented in Python. The main numerical libraries used were NumPy for simulation and vectorized computation, pandas for data handling, matplotlib for visualization, and yfinance for downloading historical SPY price data.

The implementation was organized into modular components: payoff functions, Brownian Motion and Geometric Brownian Motion simulation, Black–Scholes analytical pricing, Monte Carlo pricing, Greeks calculation, historical volatility calibration, implied volatility extraction, and model risk diagnostics.

The full Python implementation is maintained separately in the project repository as:

[spy-option-pricing-project.py](#)

Component	Python Implementation
Payoff calculation	<code>call_payoff()</code> , <code>put_payoff()</code>
GBM simulation	<code>simulate_gbm_paths()</code>
Black–Scholes pricing	<code>black_scholes_call()</code> , <code>black_scholes_put()</code>
Monte Carlo pricing	<code>monte_carlo_call_price()</code>
Greeks	<code>delta</code> , <code>gamma</code> , <code>vega</code> , <code>theta</code> , <code>rho</code>
Volatility calibration	Historical log-return based estimator
Implied volatility	Bisection root-finding method